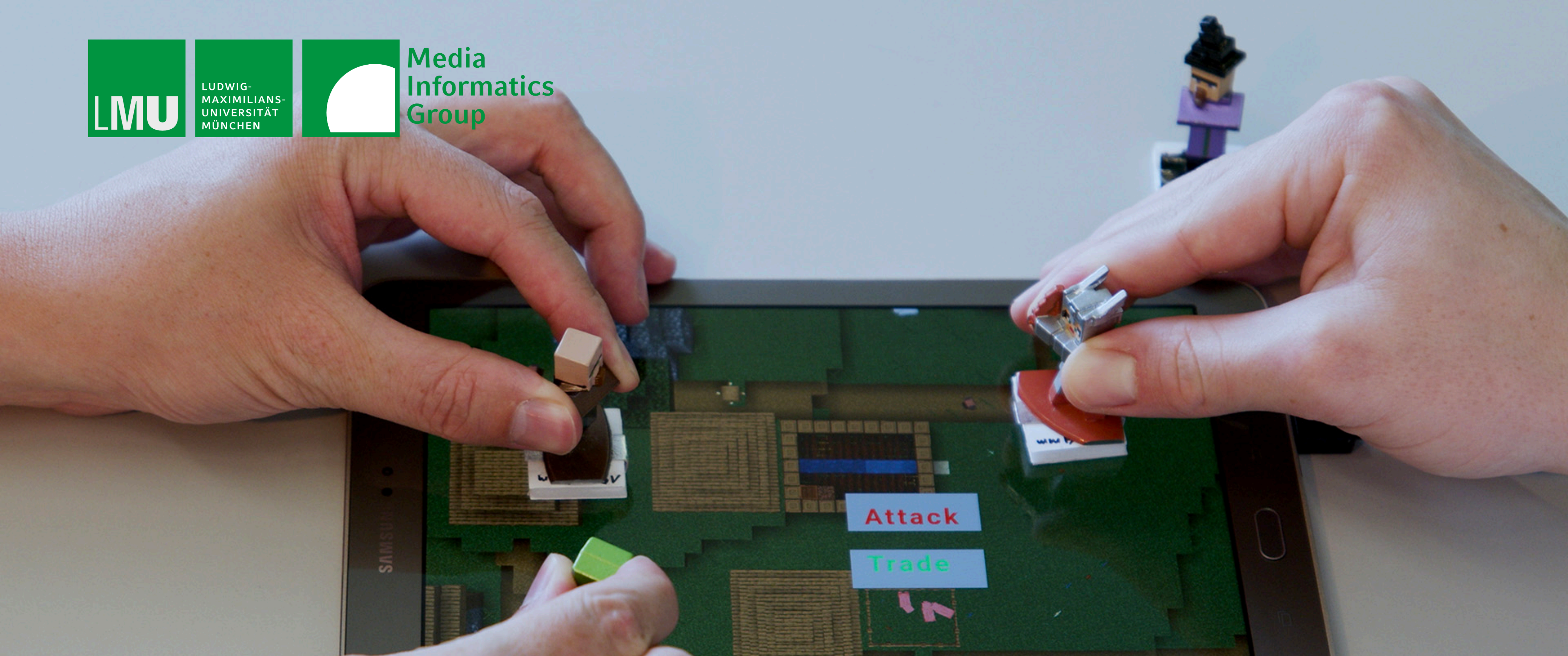
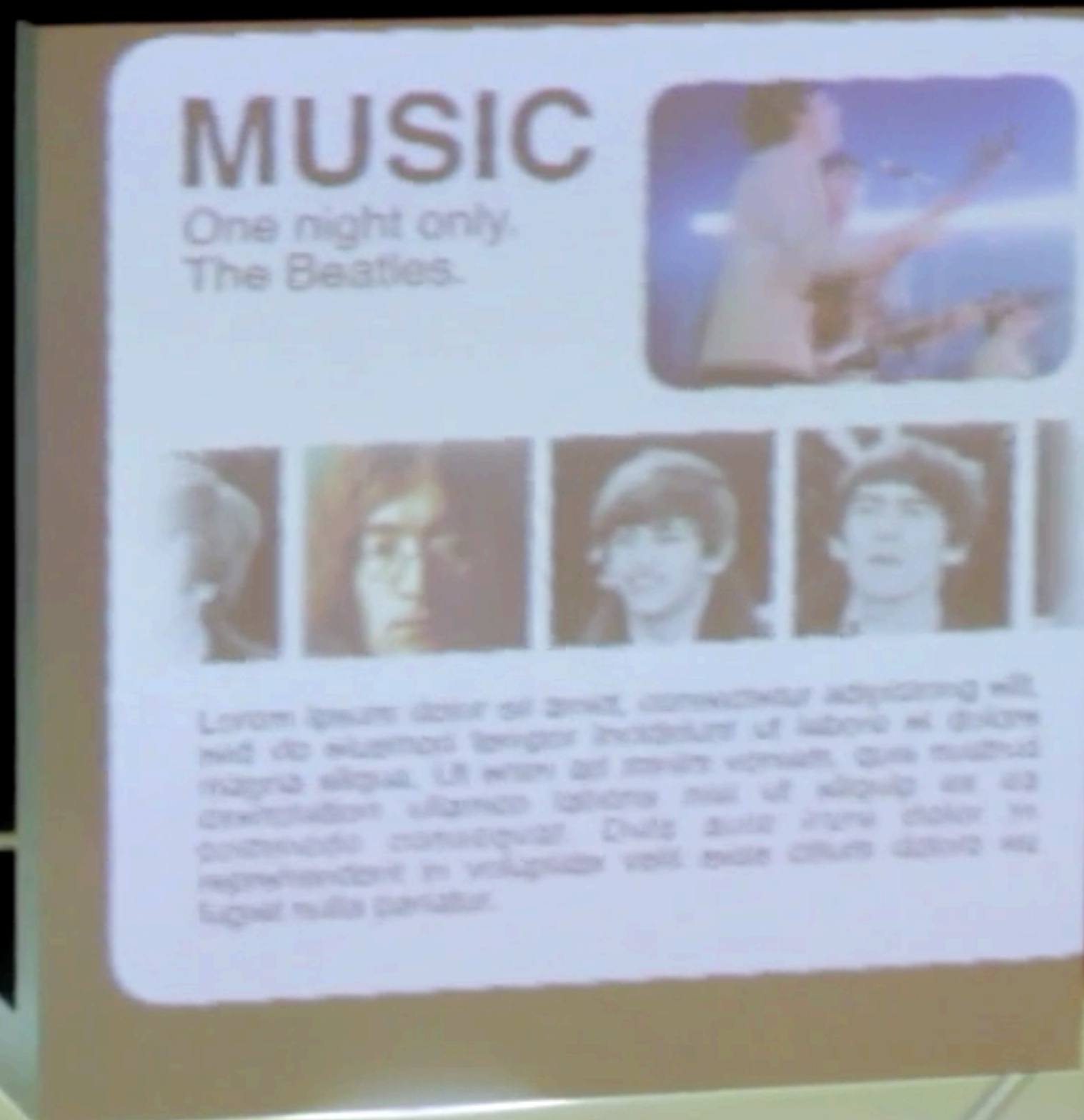
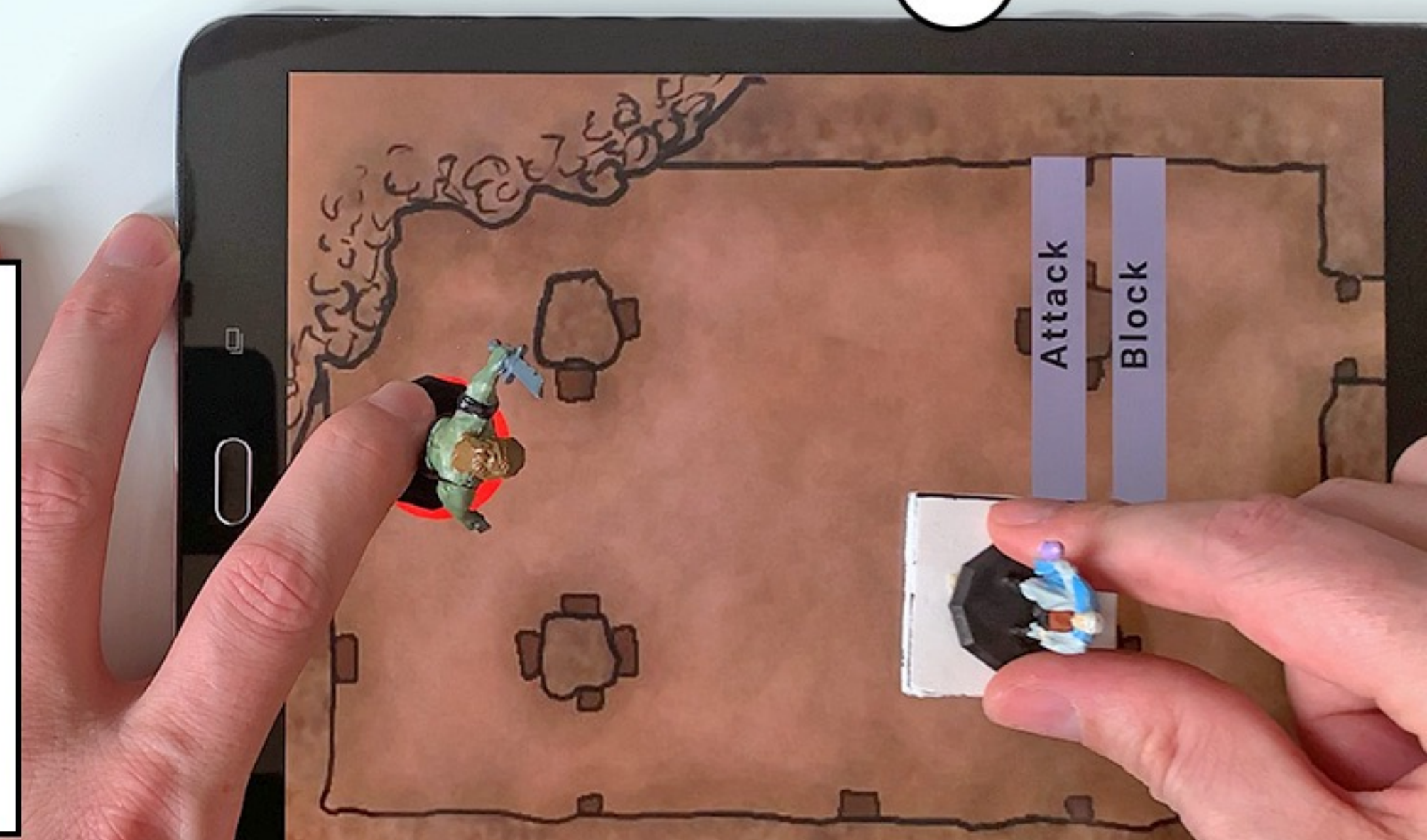
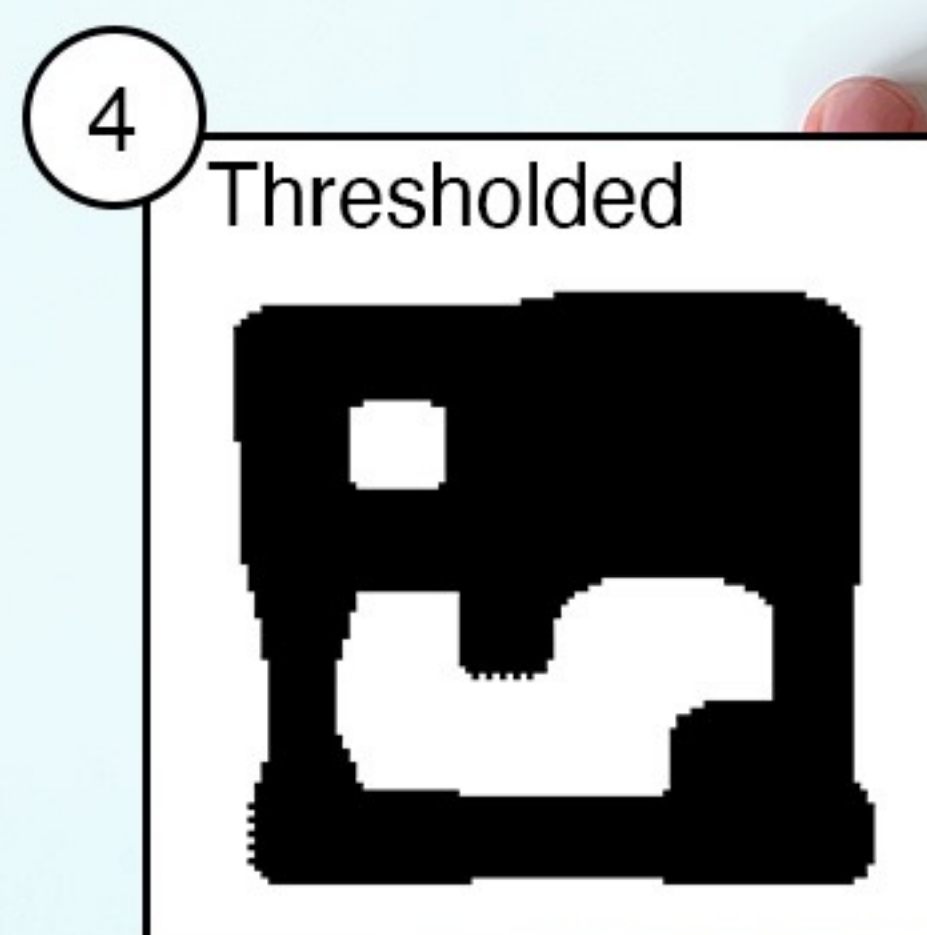
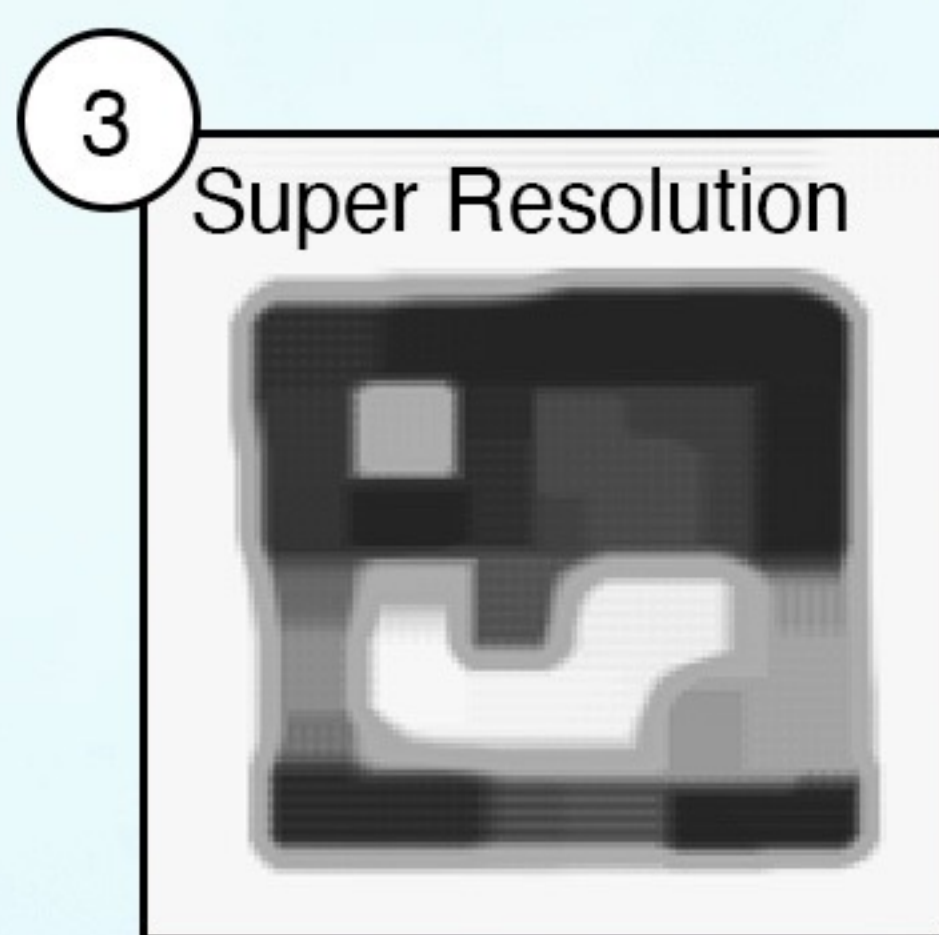
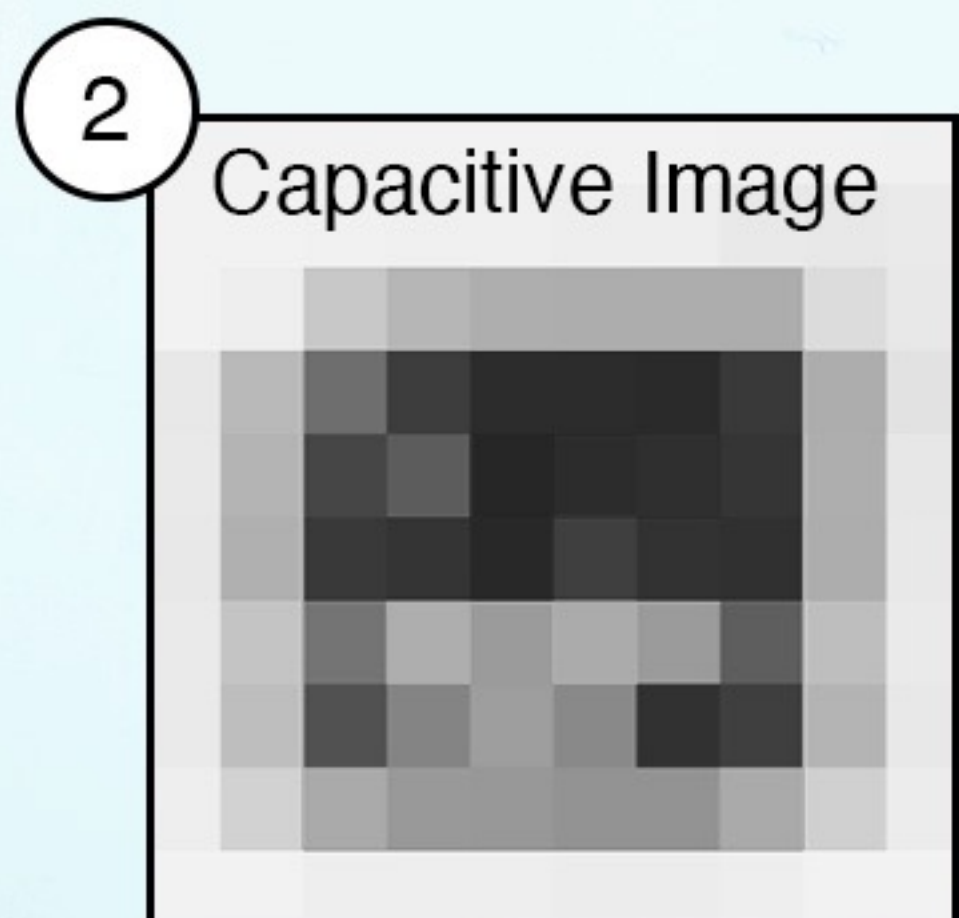




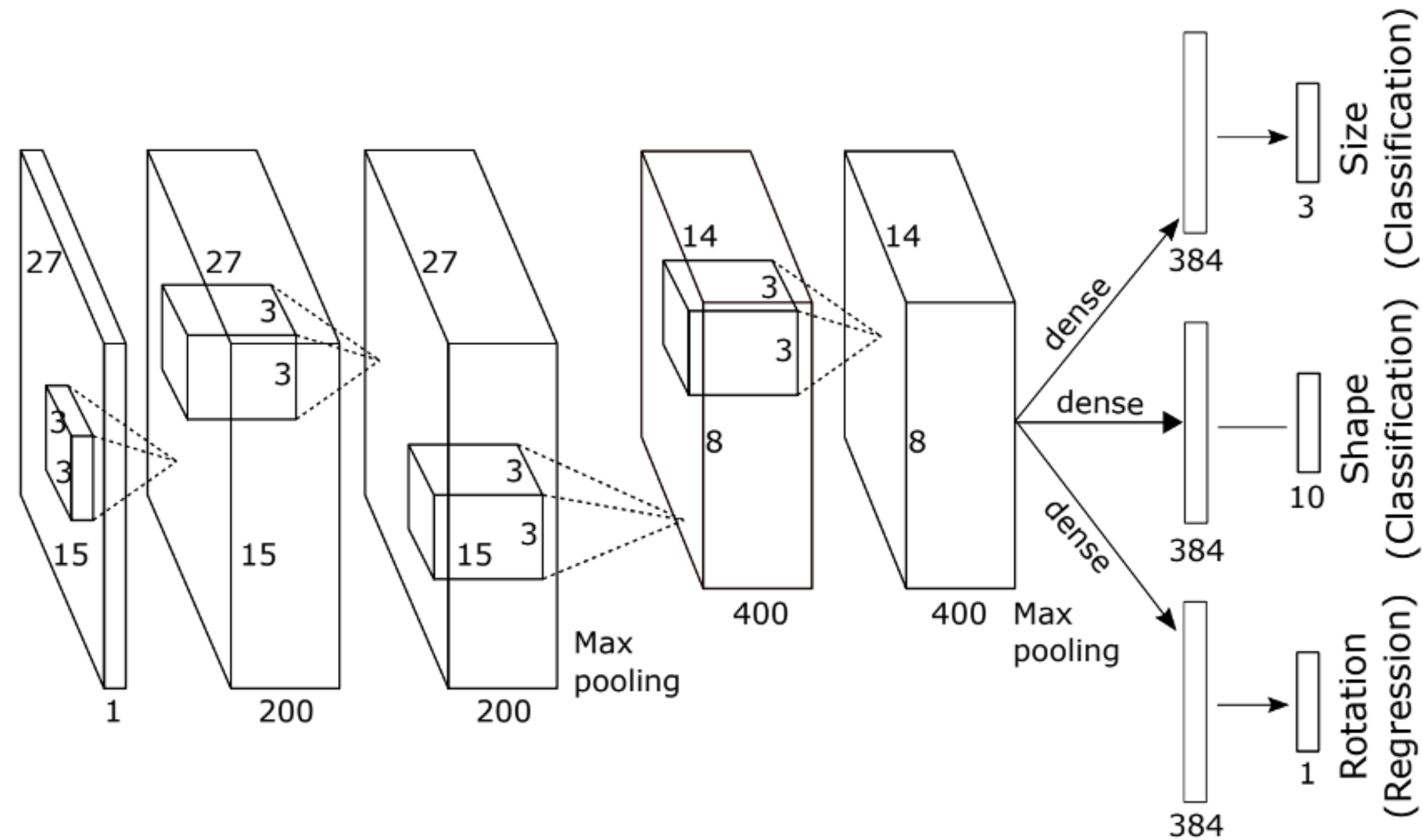
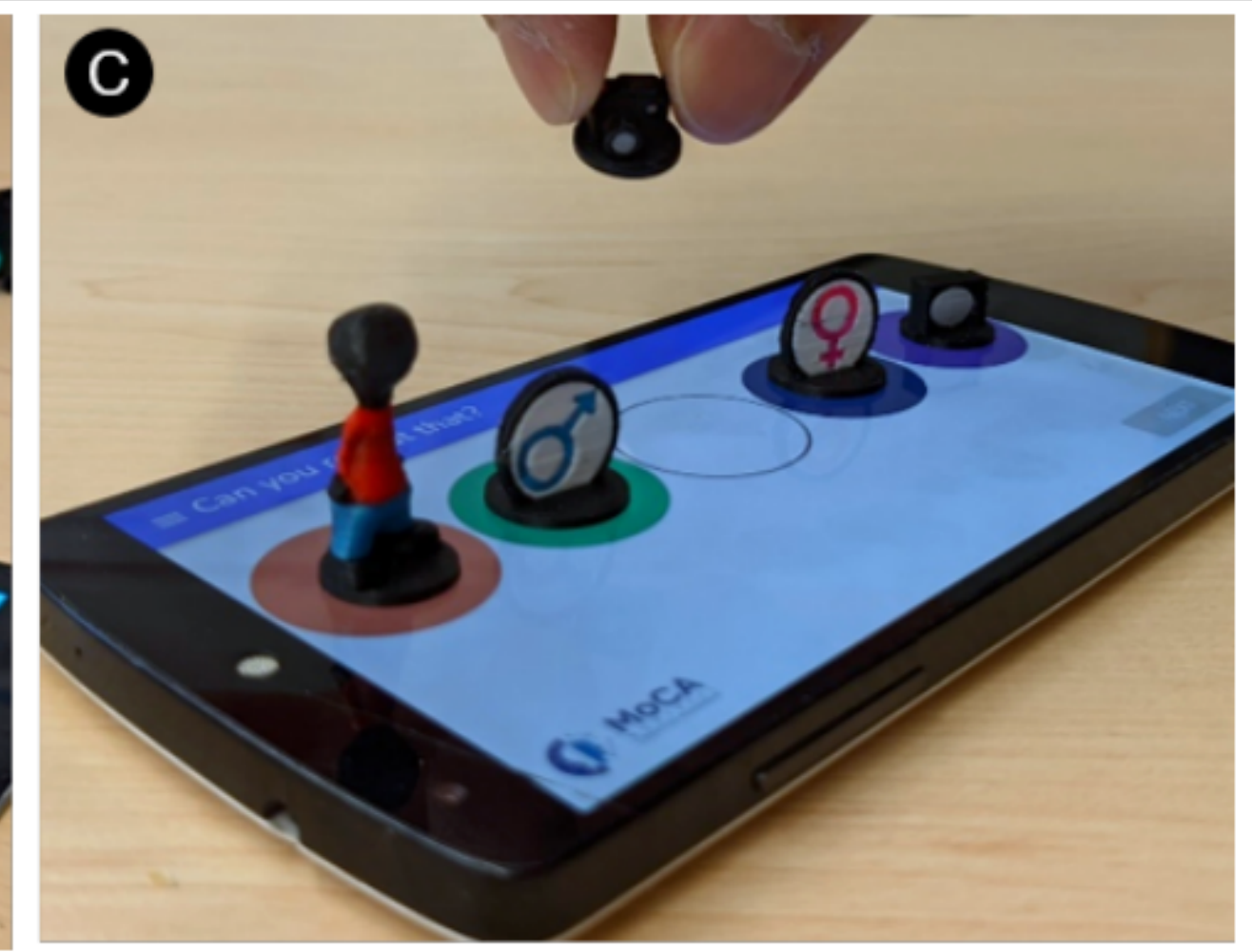
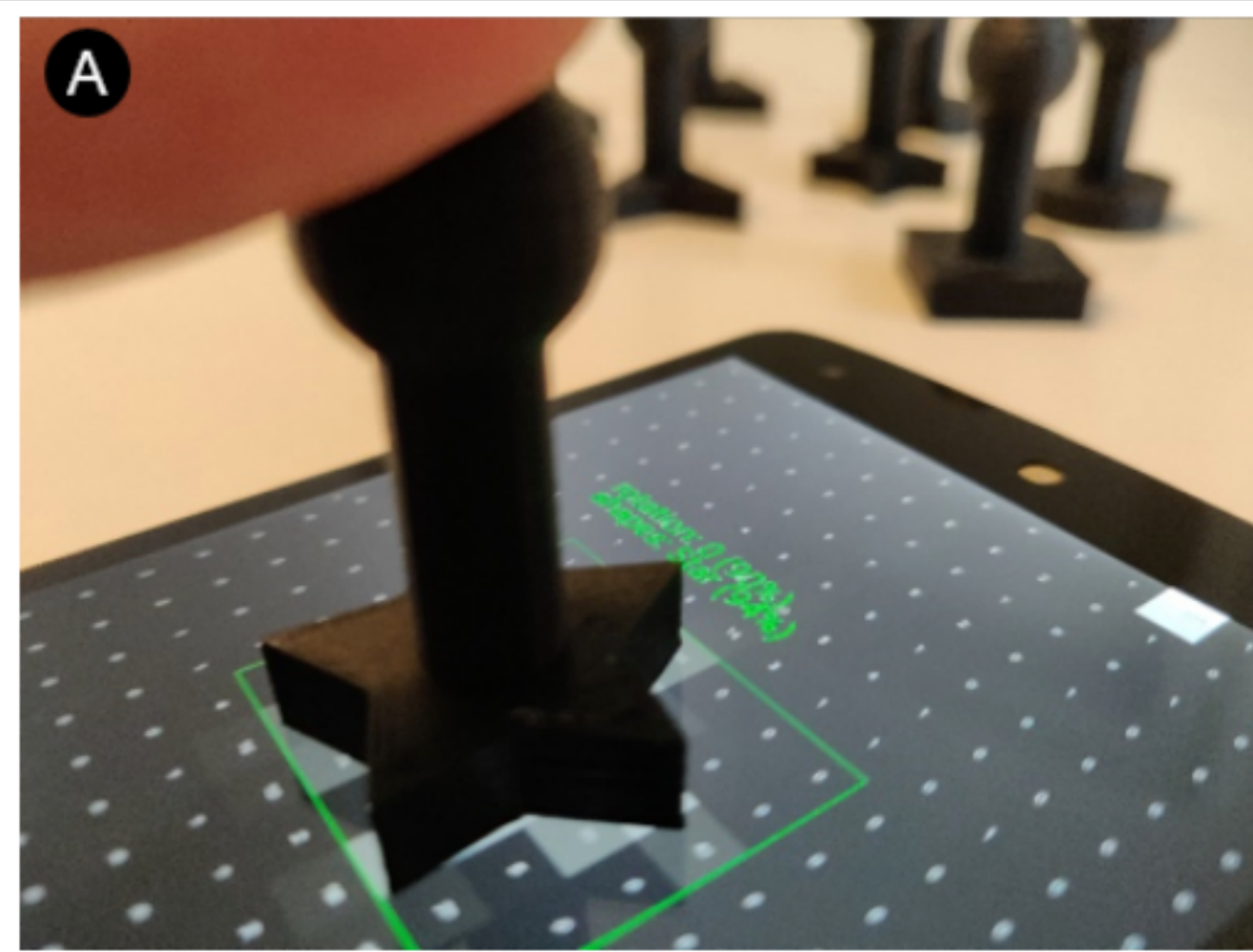
Conductive Fiducial Tangibles for Everyone: A Data Simulation-Based Toolkit using Deep Learning

Benedict Steuerlein and Sven Mayer









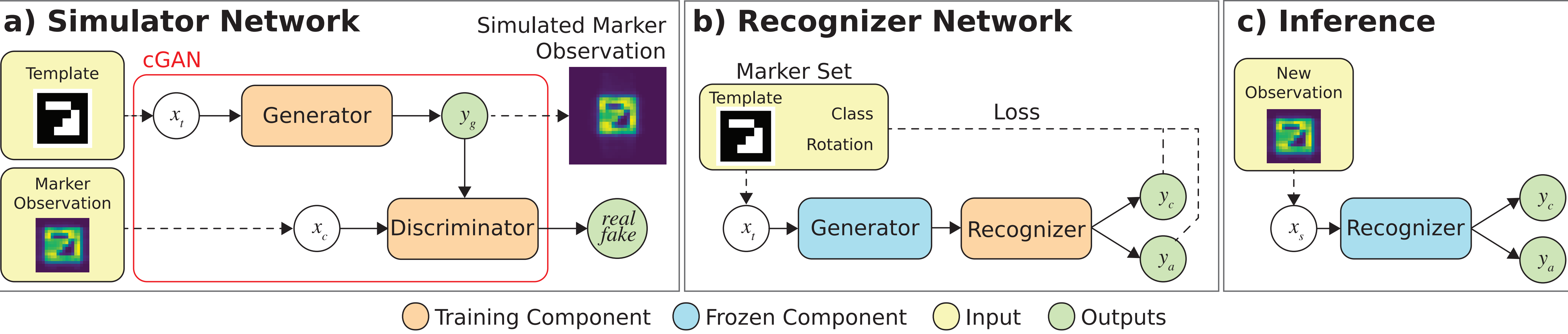
Drawbacks with Current Solutions

Current Geometric Super-Resolution Approaches are Slow

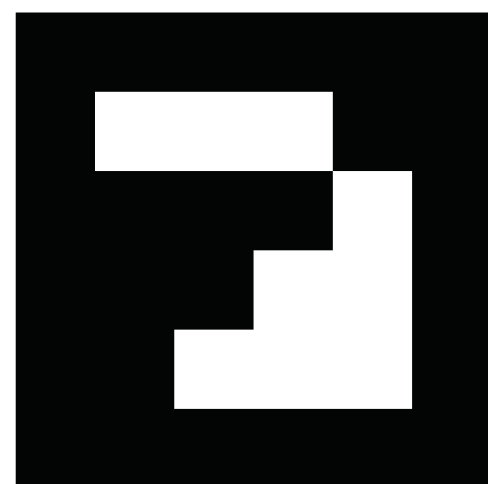
Deep Learning Approaches are Trained on Pre-Define Markers

Deep Learning Approaches Require Domain Knowledge for Training

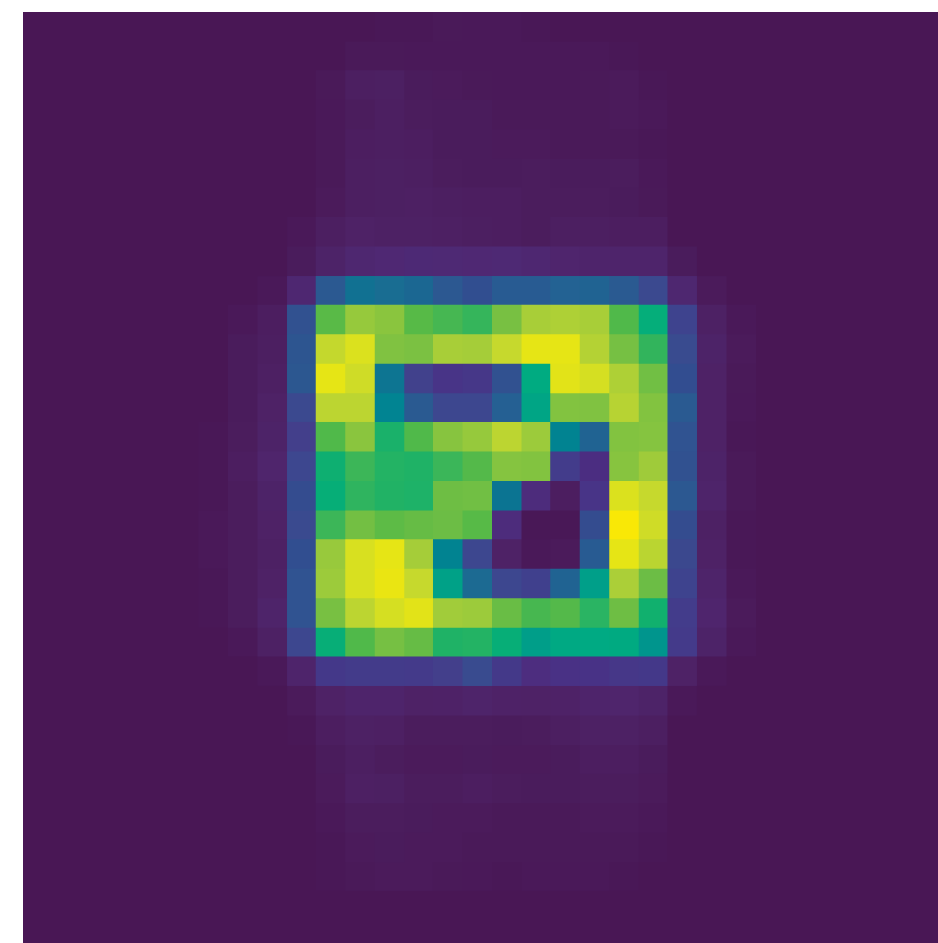
Current Approaches are Cumbersome to Use



Our Approach

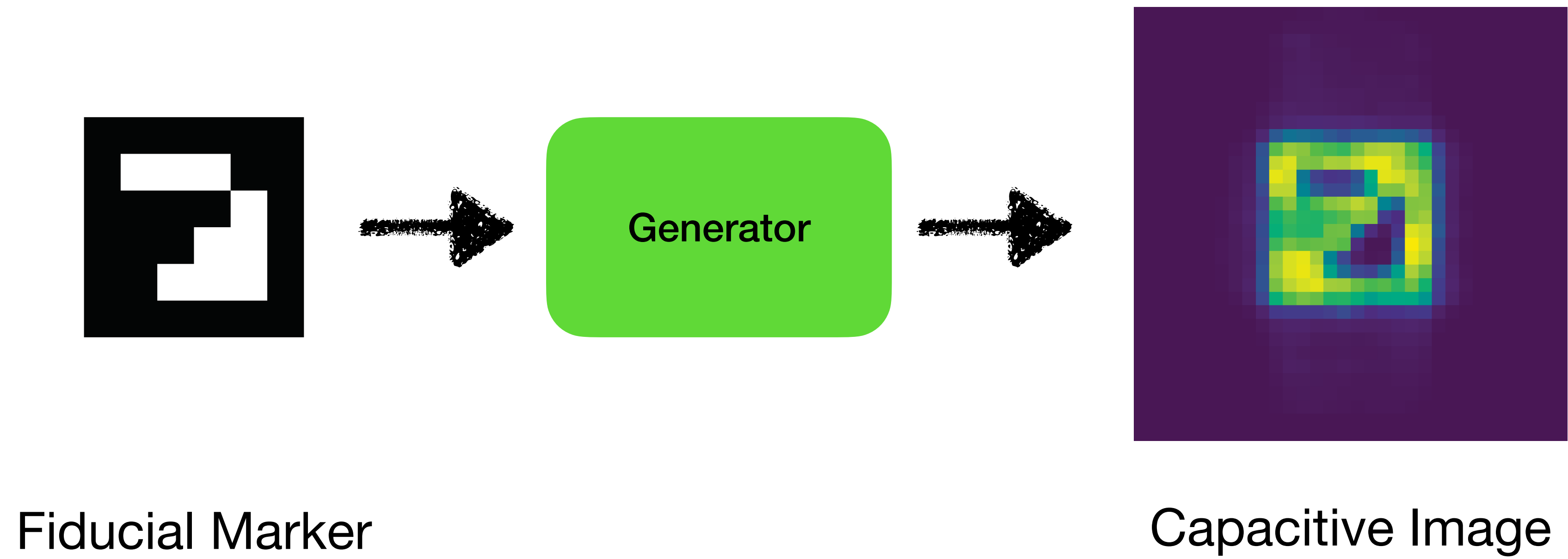


Fiducial Marker

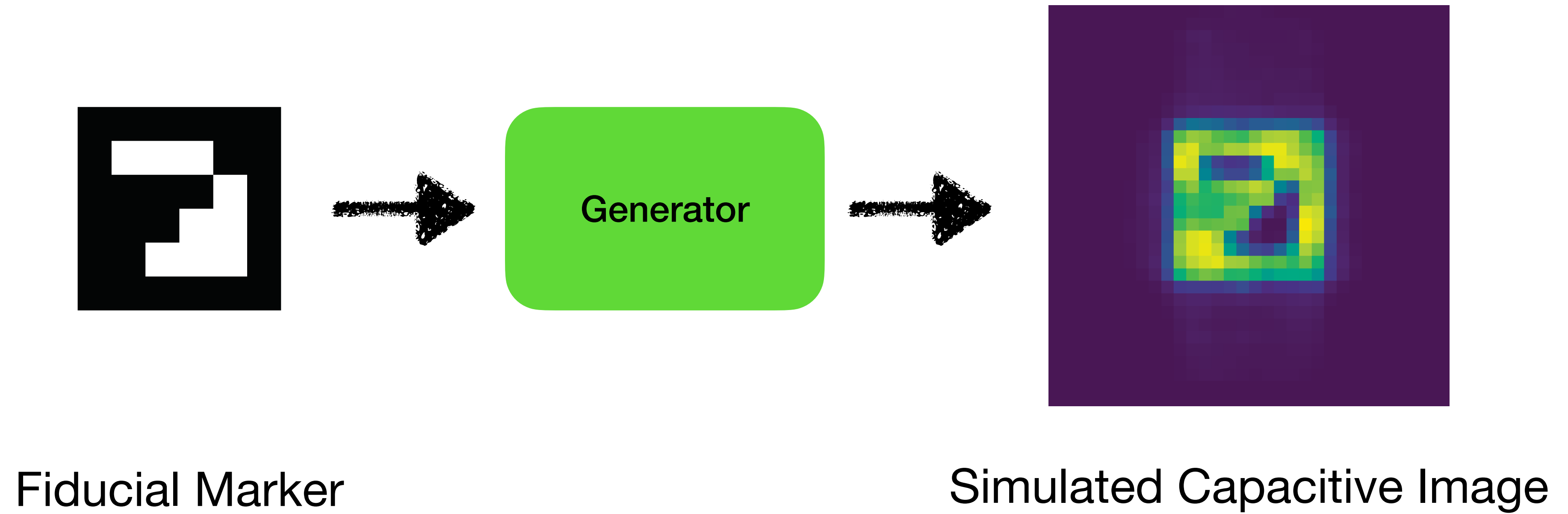


Capacitive Image

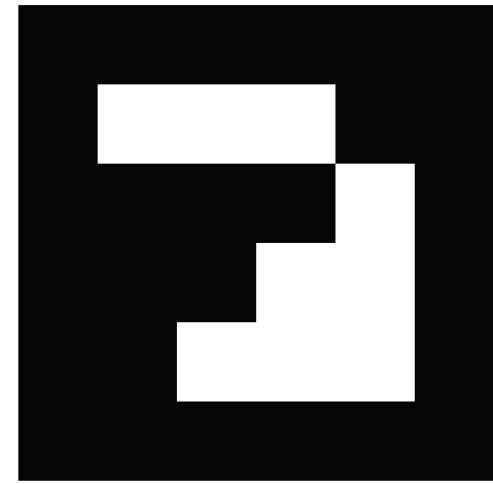
Our Approach



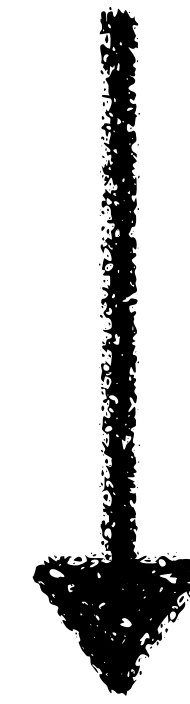
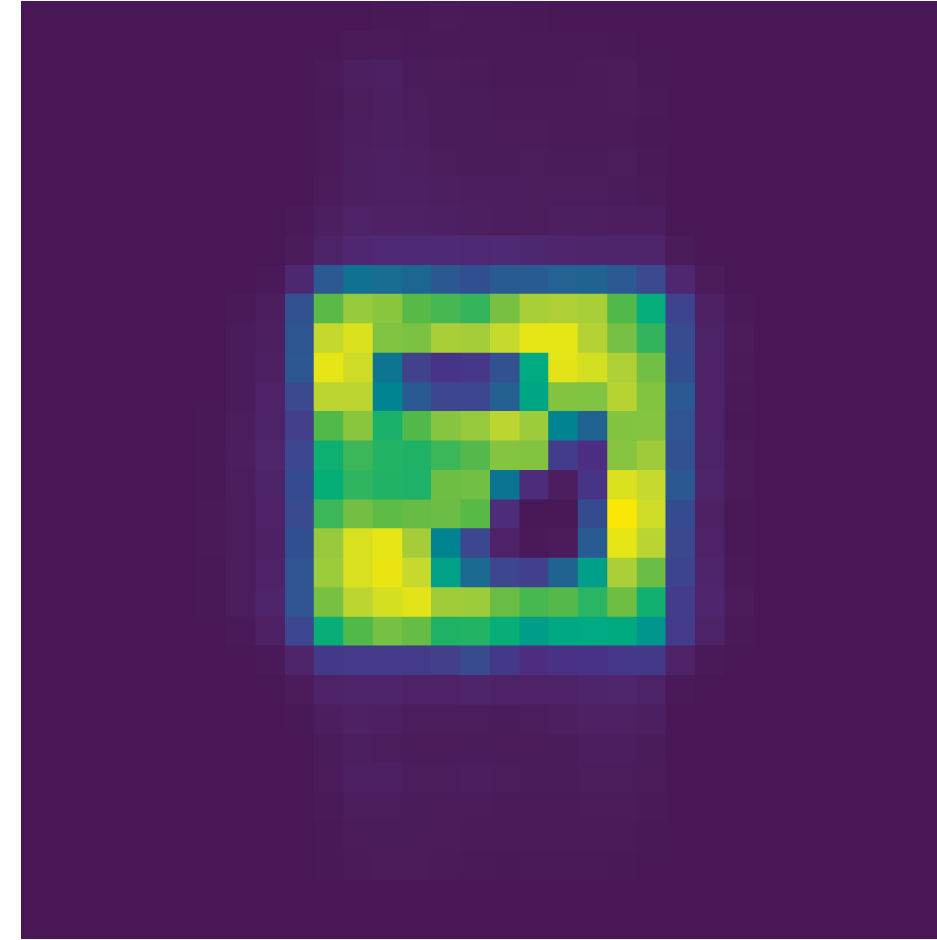
Our Approach



Our Approach



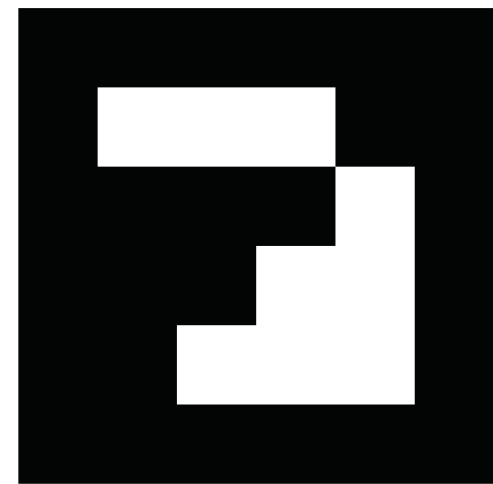
Generator



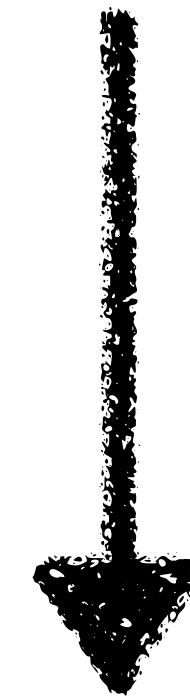
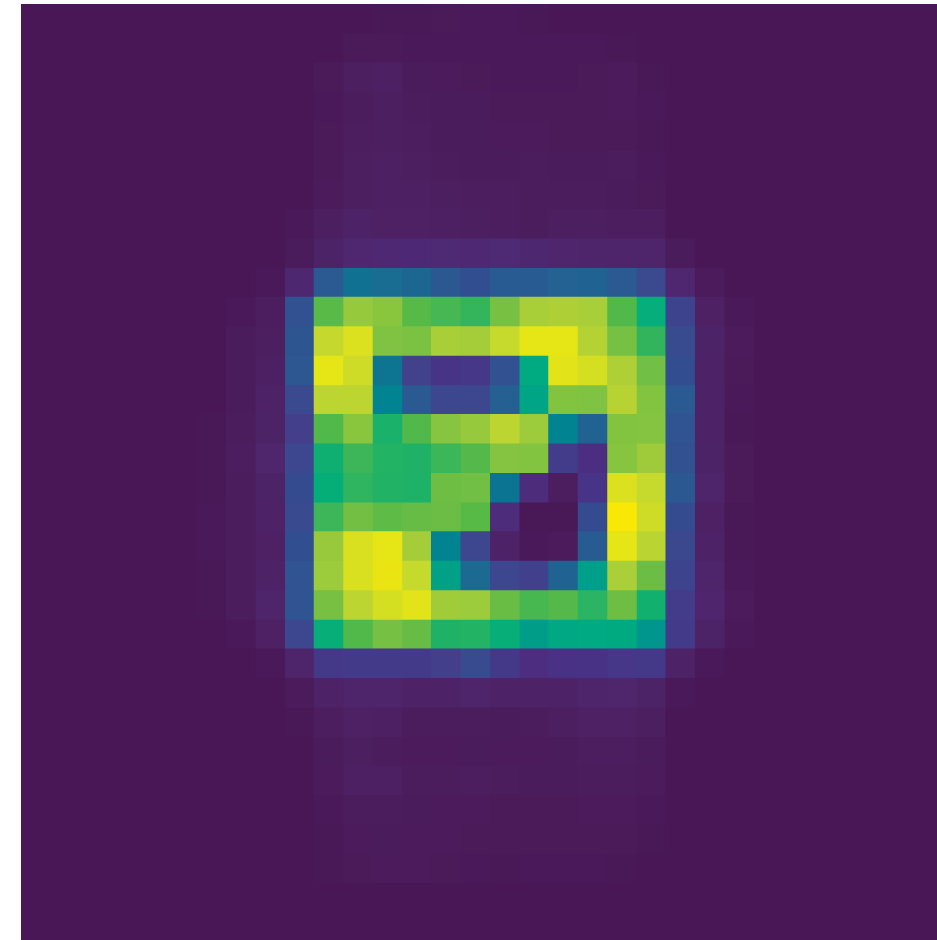
Training using simulated data

Recognizer

Our Approach

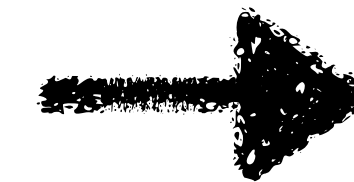


Generator



Training using simulated data

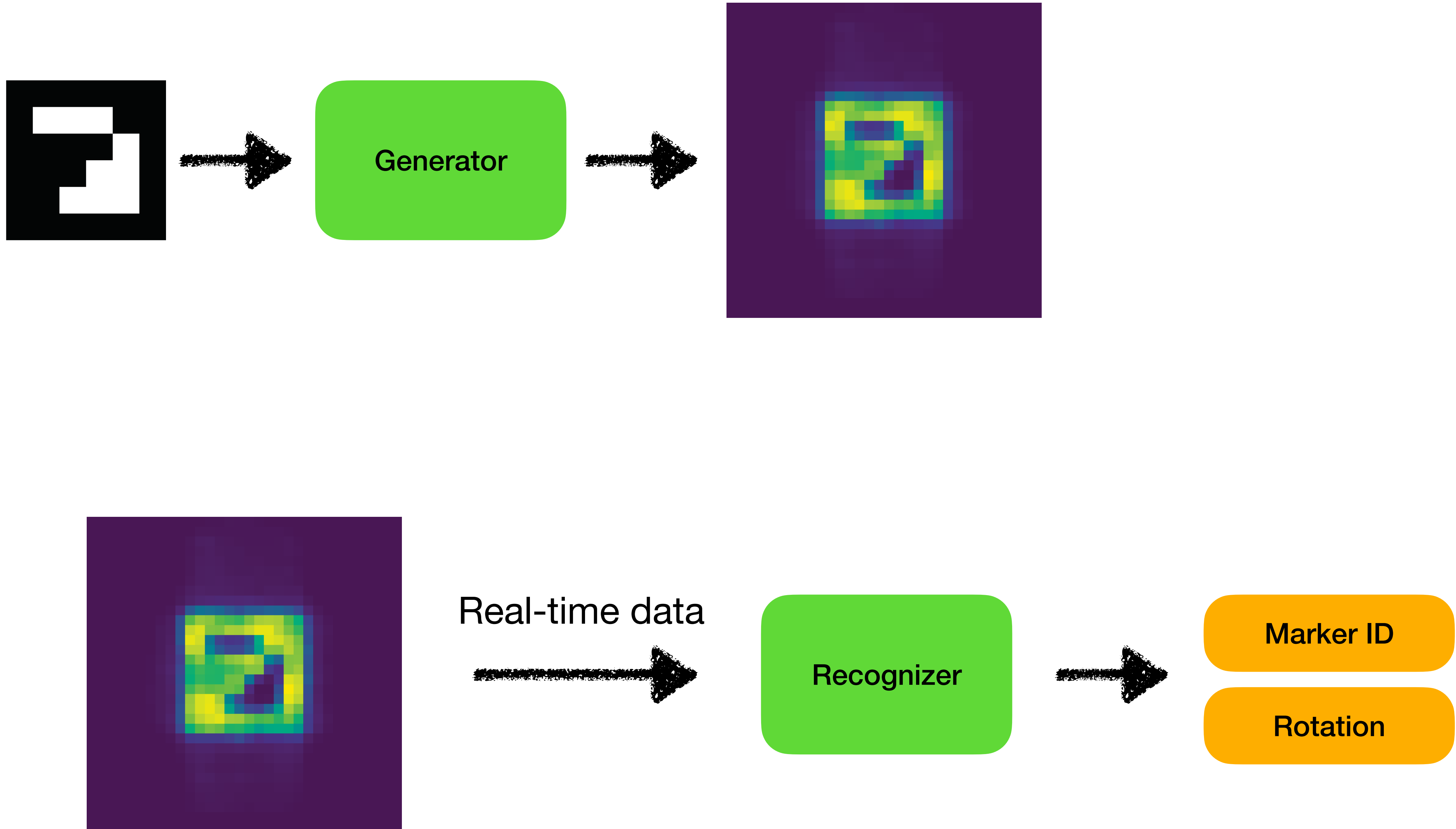
Recognizer



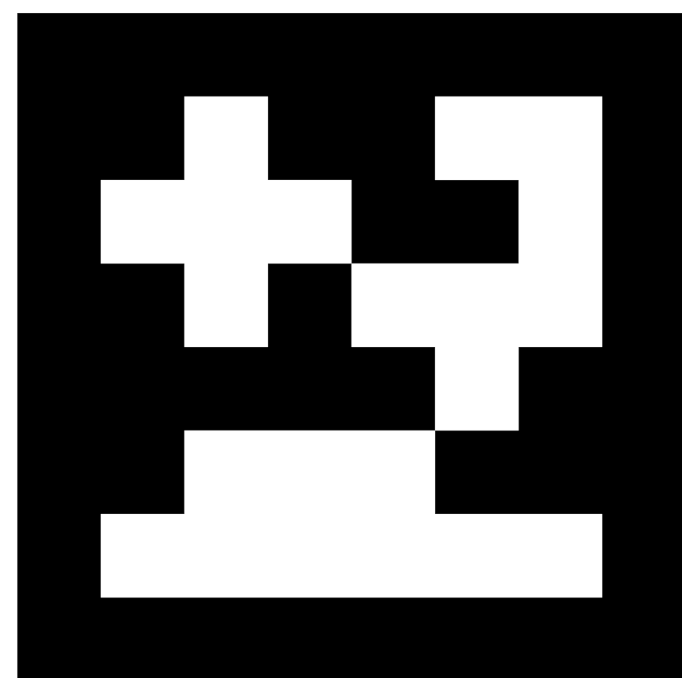
Marker ID

Rotation

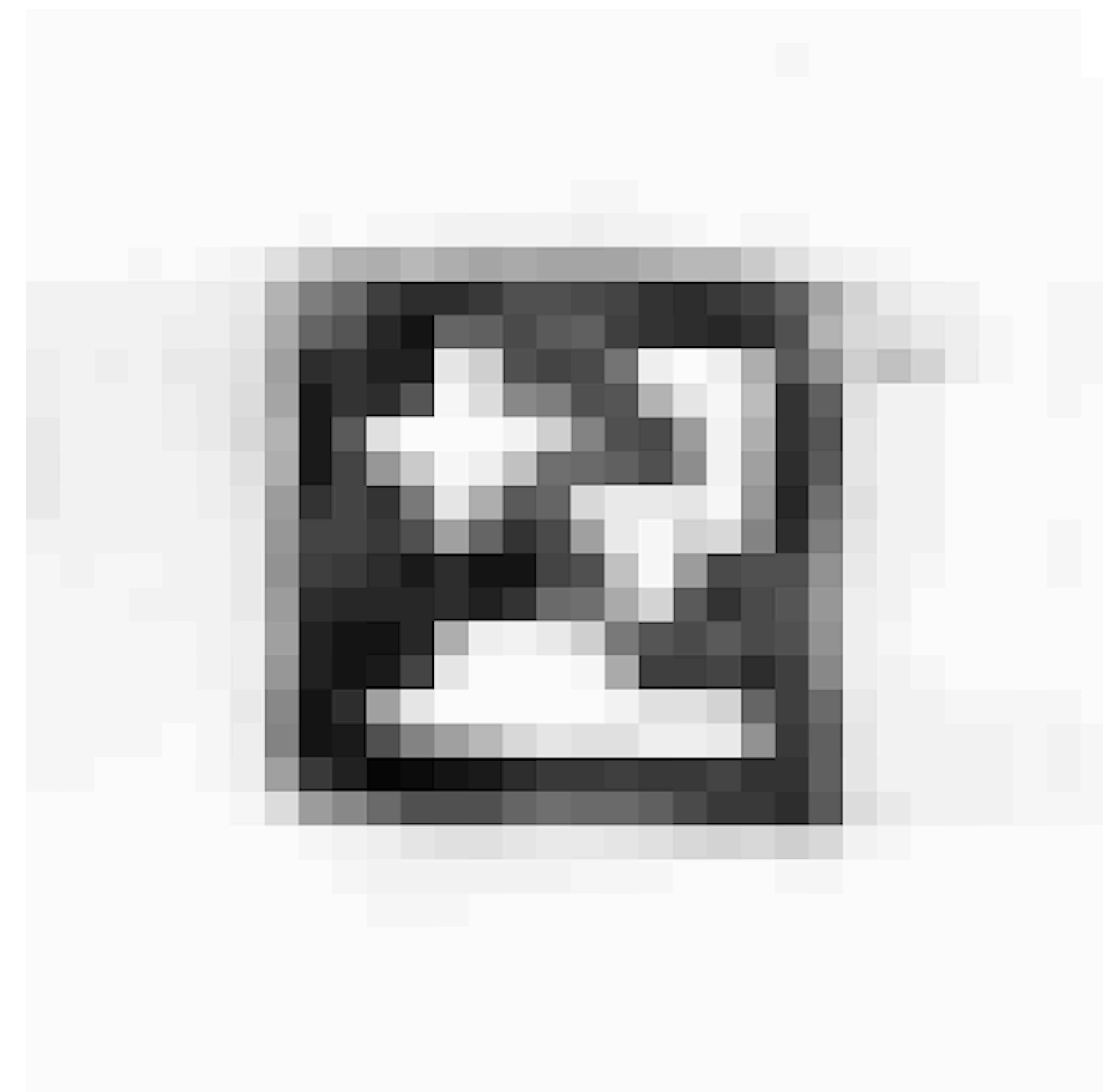
Our Approach



Capacitive Simulator Network

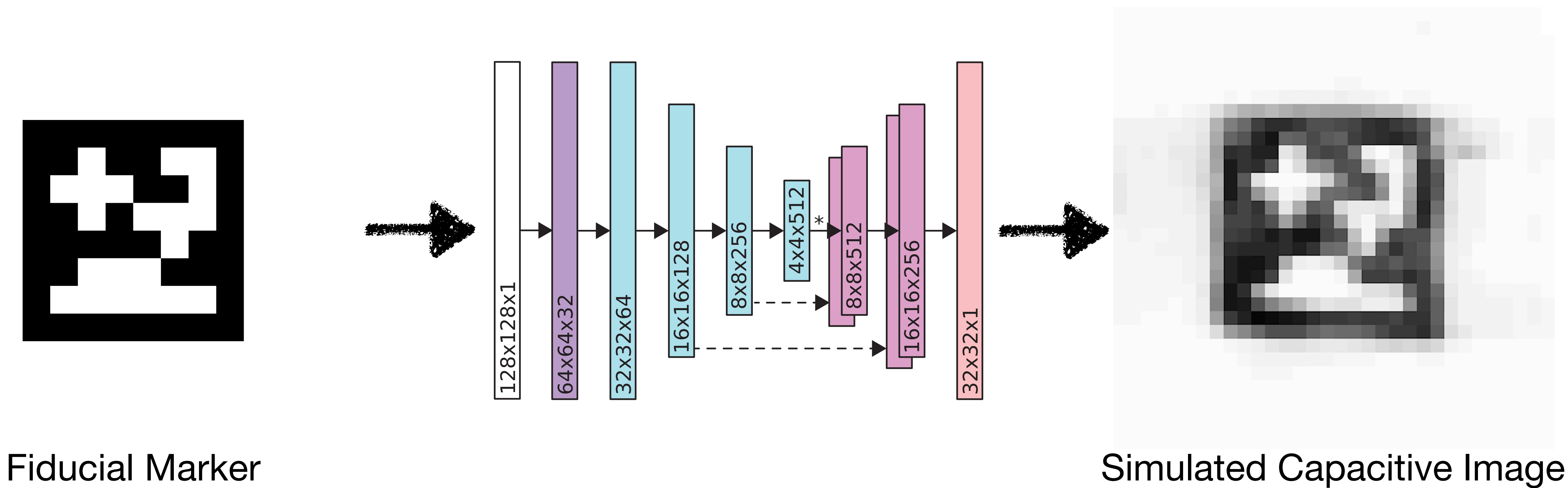


Fiducial Marker

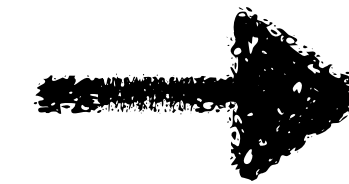
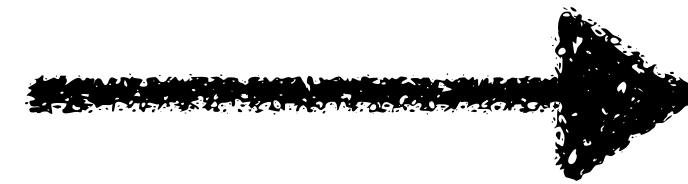
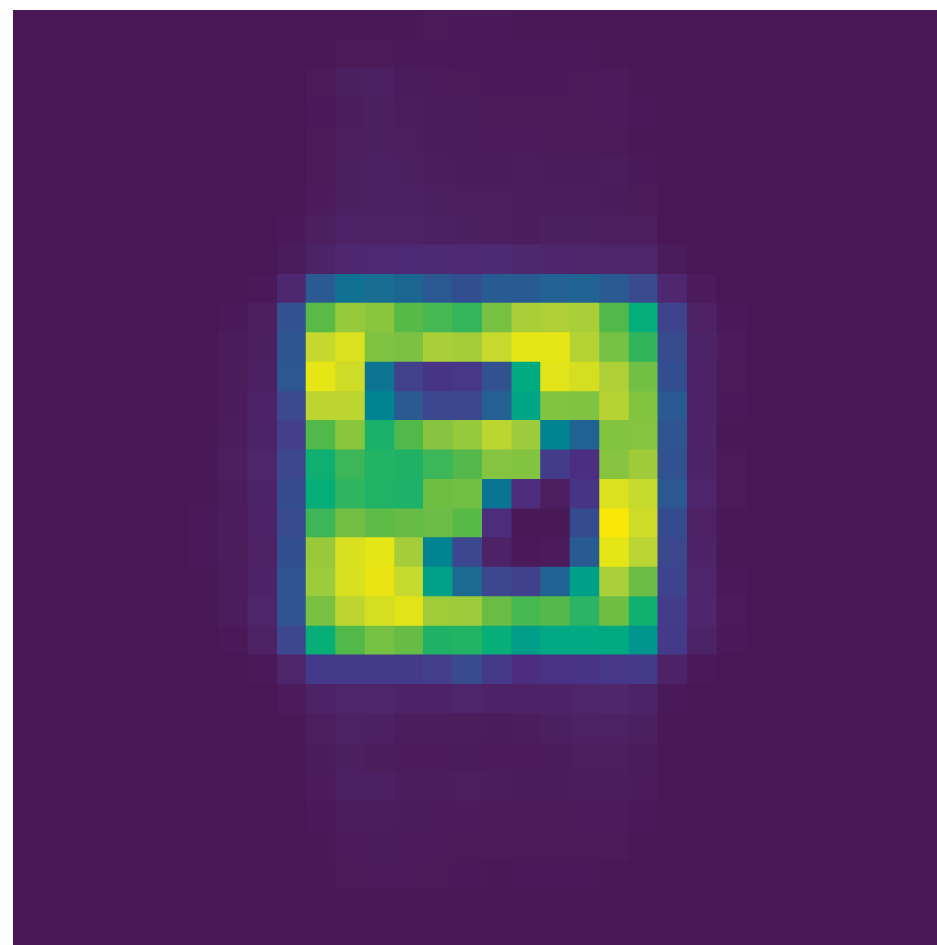


Simulated Capacitive Image

Capacitive Simulator Network



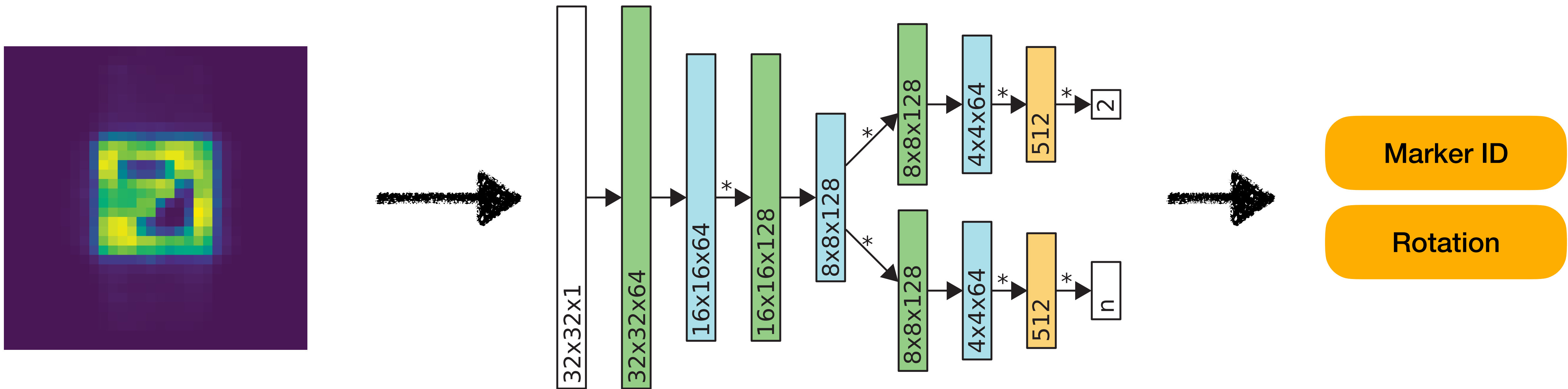
Object Recognition Network



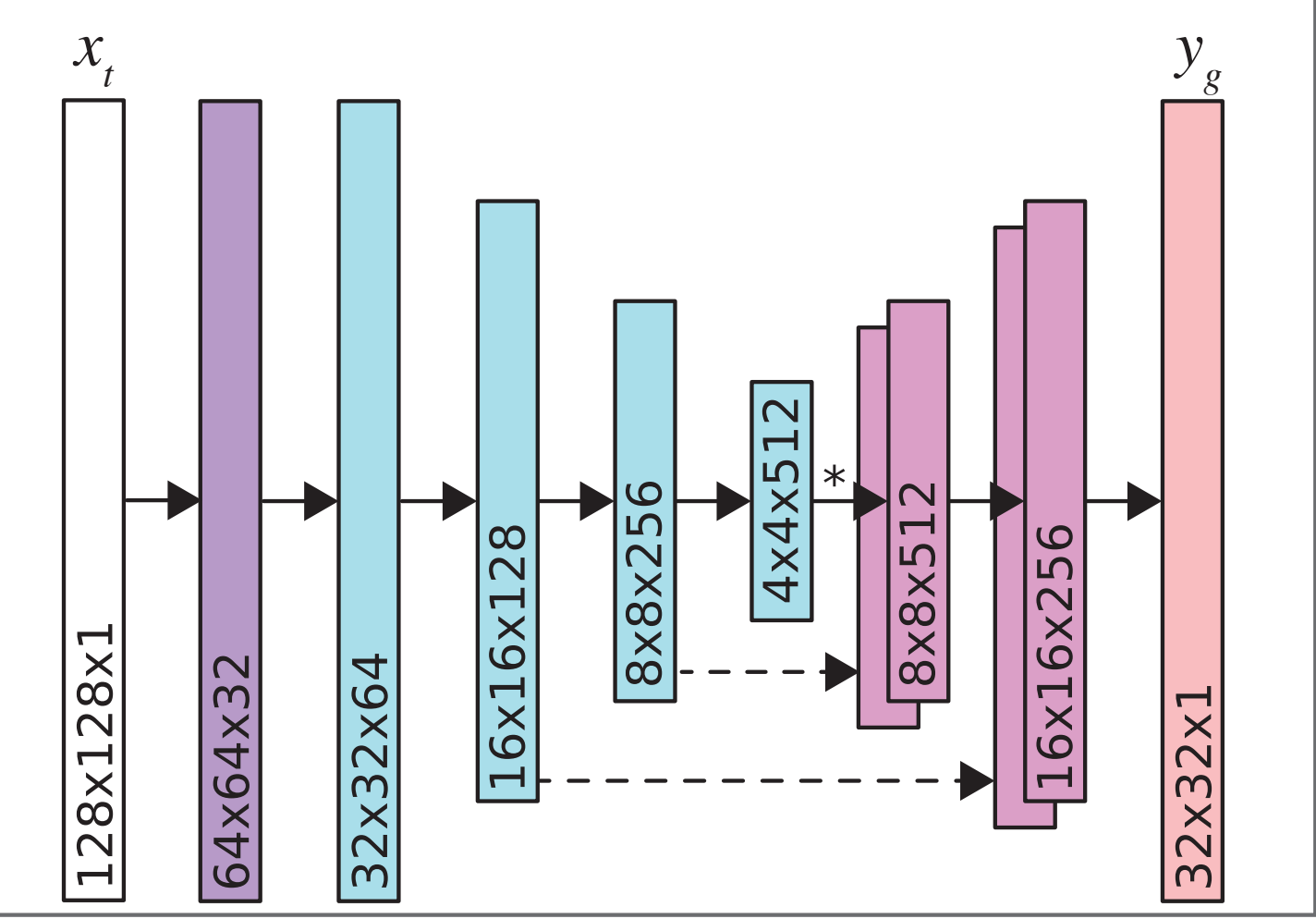
Marker ID

Rotation

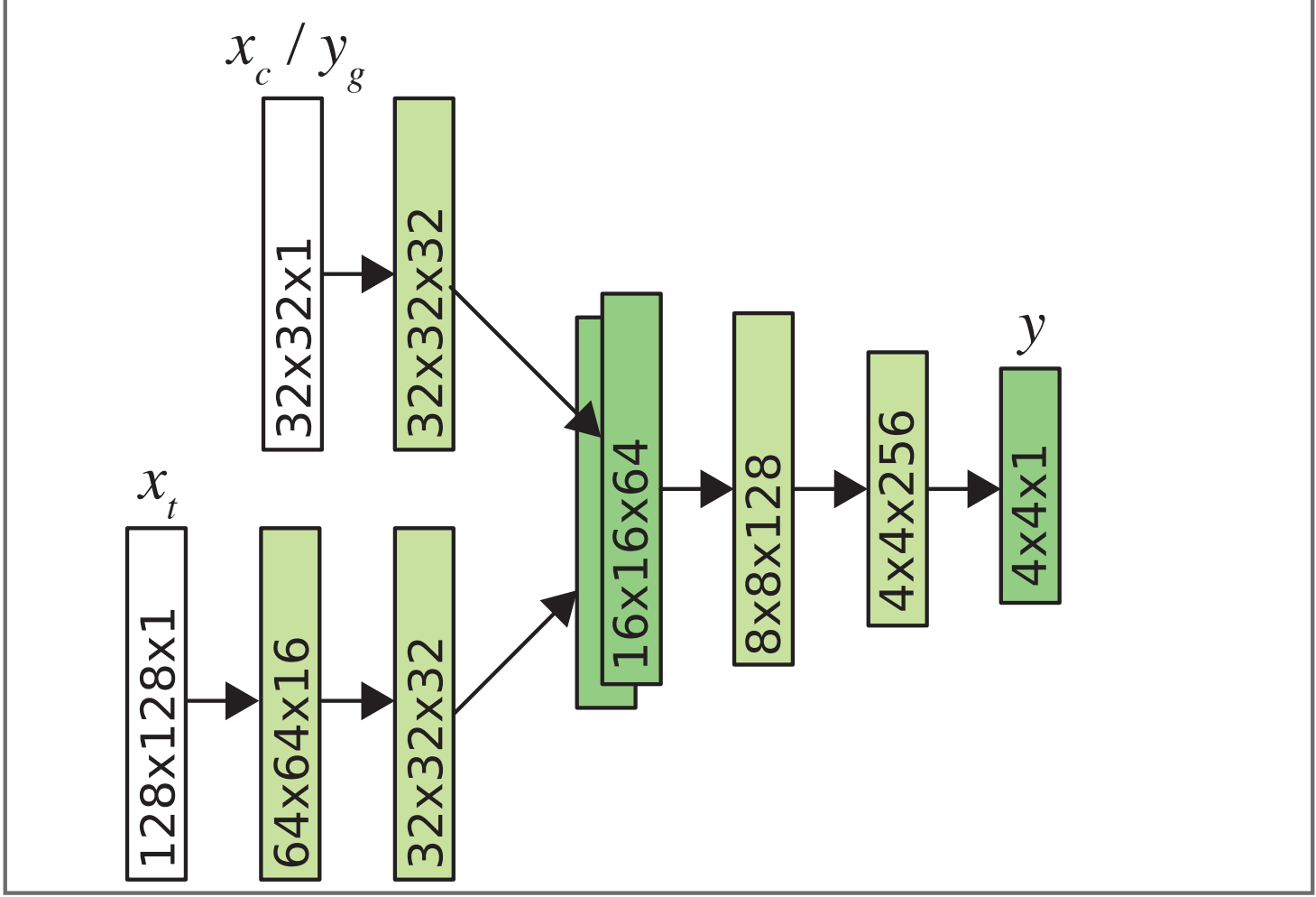
Object Recognition Network



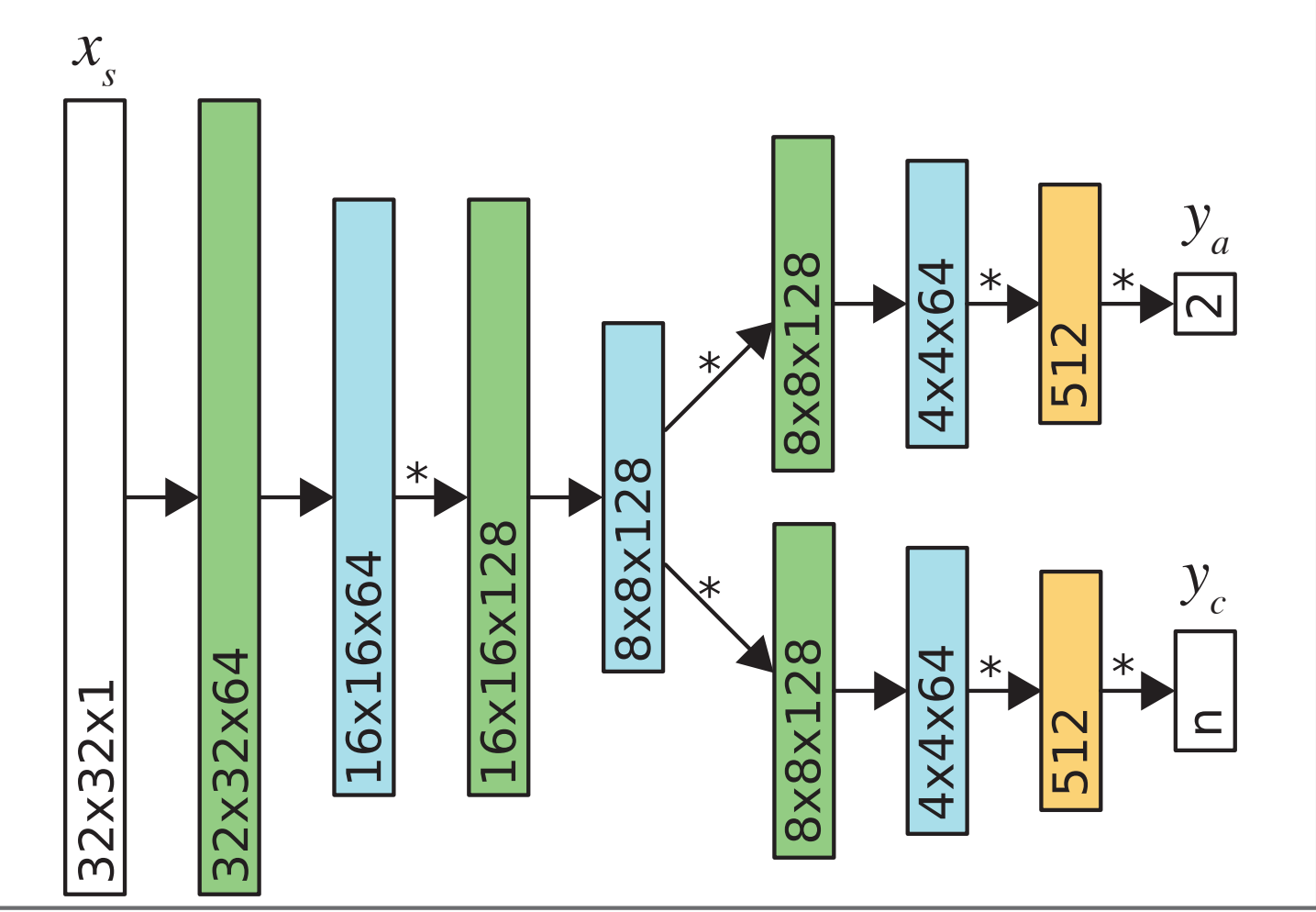
a) Generator Network



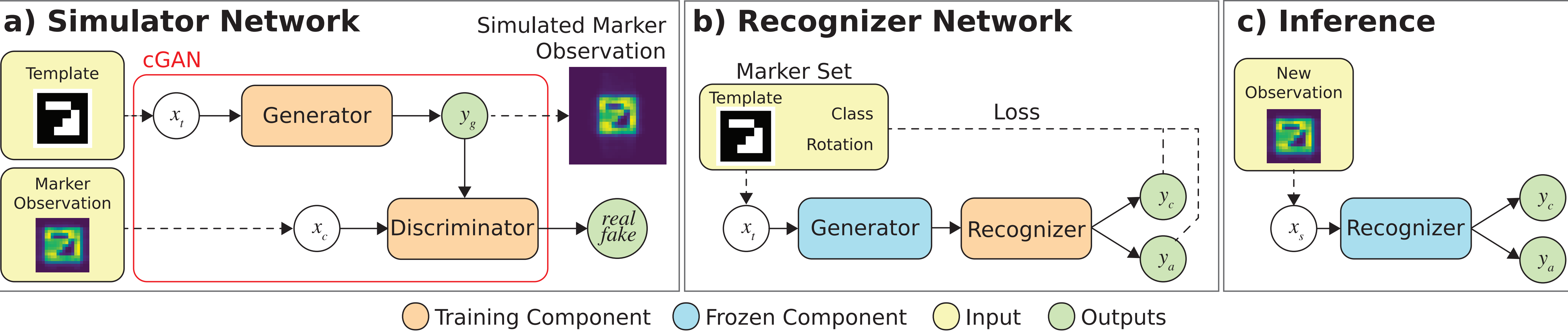
b) Discriminator Network



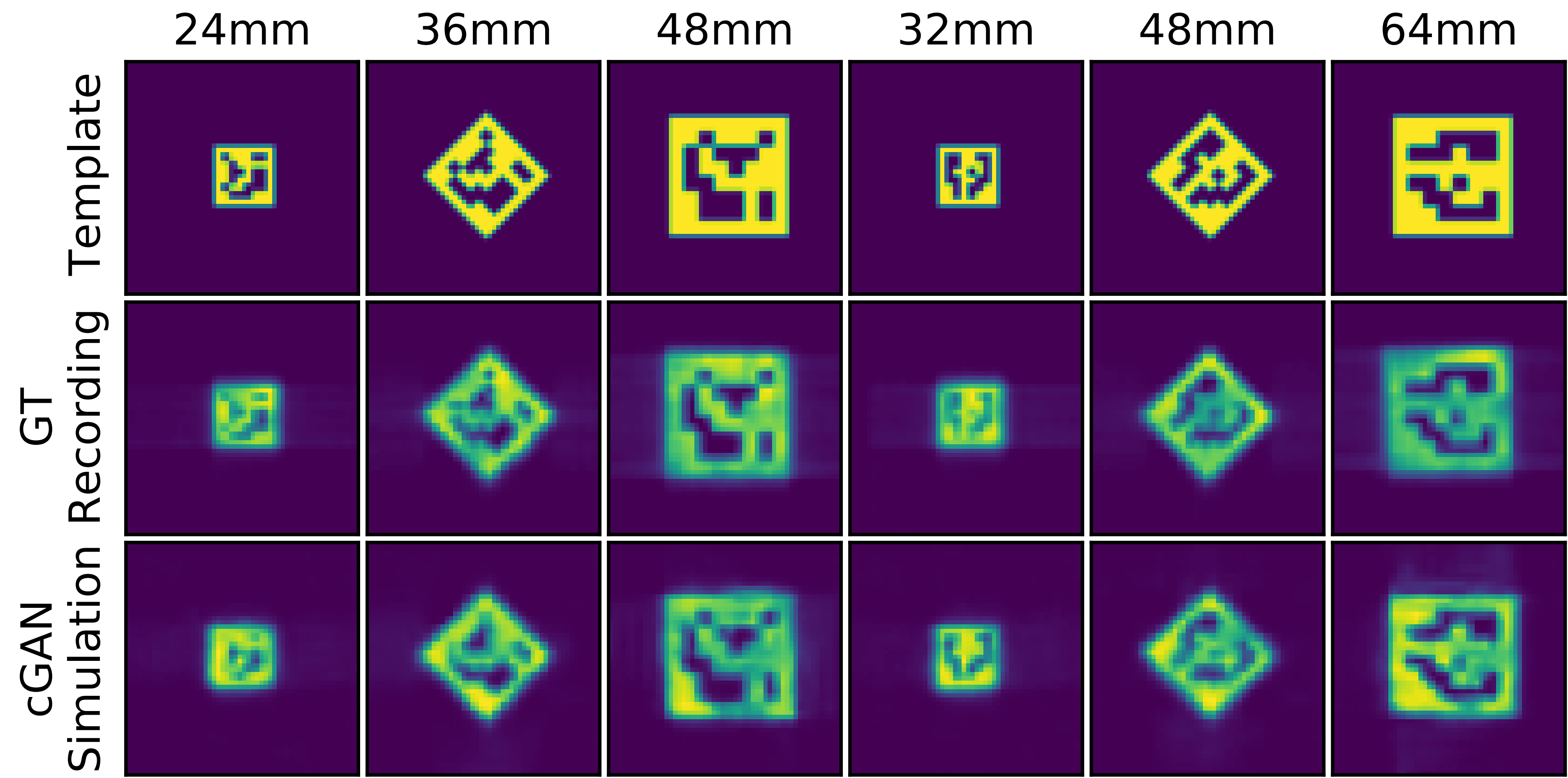
c) Recognizer Network

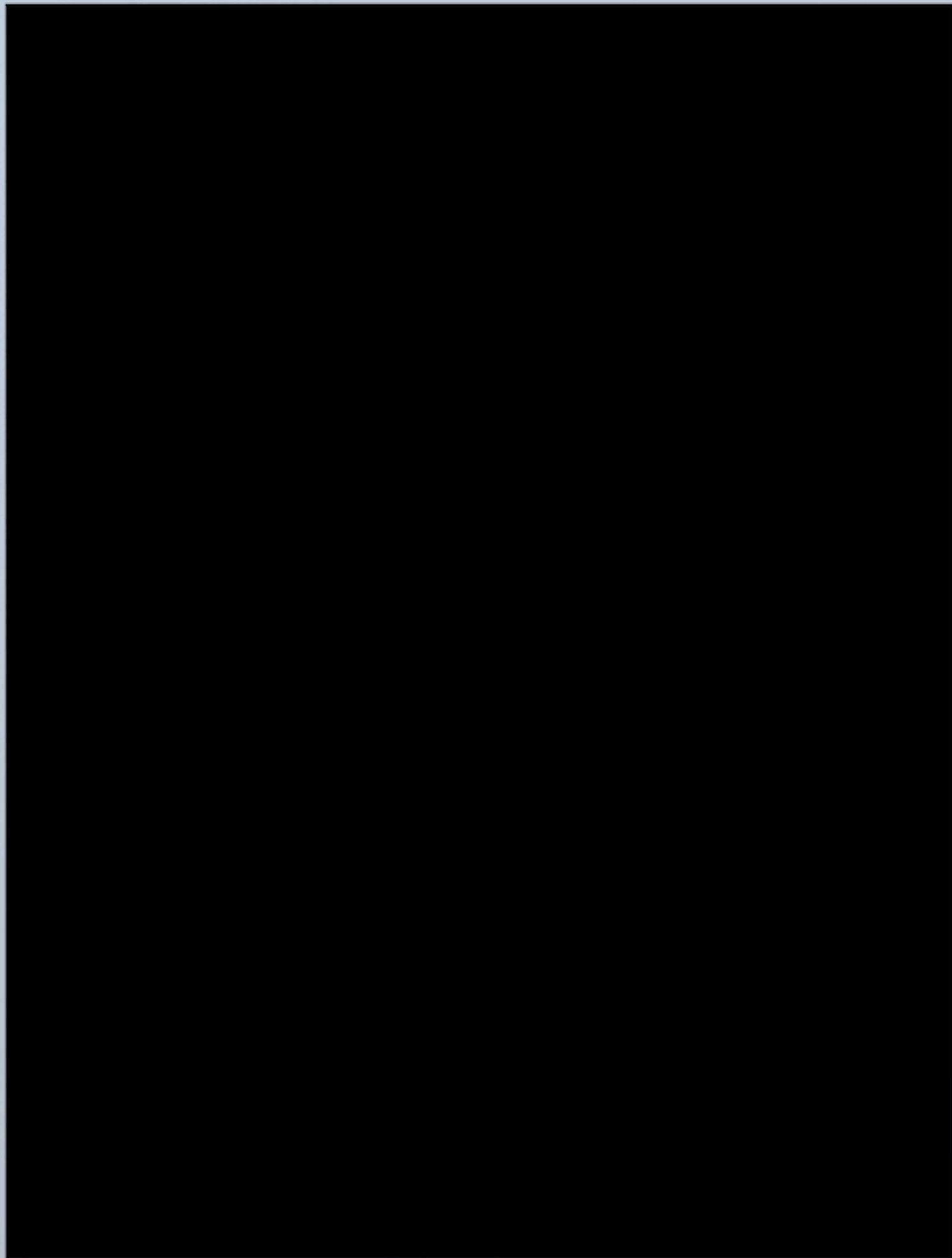


- Input/Output
- Dense + BatchNorm
- CNN
- CNN + MaxPool
- CNN + BatchNorm
- CNN + BatchNorm + MaxPool
- Upsampling + CNN
- Upsampling + CNN + BatchNorm
- Concatenate
- Skip Connections
- Dropout

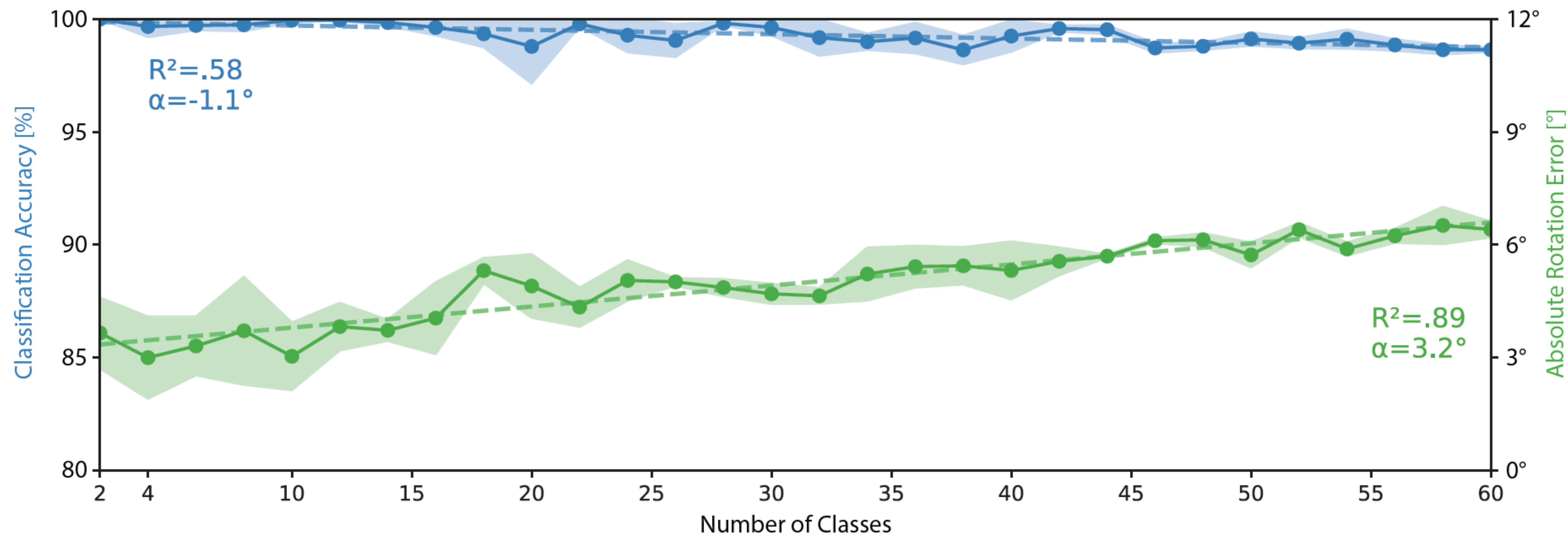


Generator Results

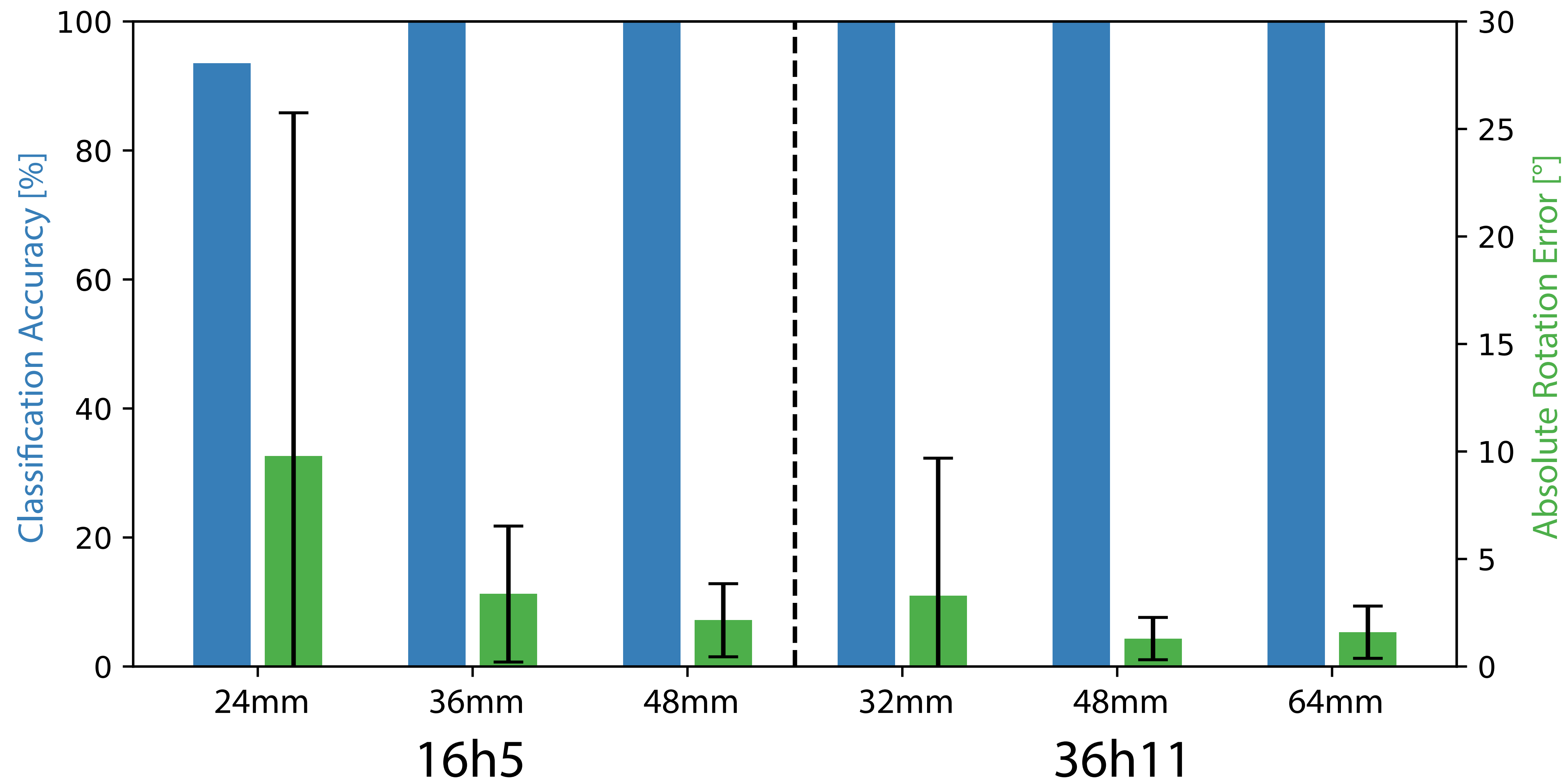




Performance



Performance



Toolkit

- ▶ Open-source toolkit
- ▶ Containing a pre-trained generator for marker simulation
- ▶ Flexible recognizer model with pre-defined hyperparameter
- ▶ Plug-and-play solution requiring footprint markers as input
- ▶ TensorFlow model to detect use-case-specific markers

<https://github.com/mimuc/Conductive-Fiducial-Marker-Simulation-Toolkit>

Conclusion

Generative Adversarial Networks for Tangible Recognition

No Expert Knowledge Required

Flexibility for Designers and Practitioners

High Accuracy Access Many Classes